

S.No	Particulars	Theory		Practical	
		Amplitude	Frequency	Amplitude	Time Frequency
1.	Message Signal	4 V	400 Hz	4 V	$T_m = x_2 - x_1$ $= 0.0025s$ 400 Hz
2.	Carrier Signal	10 V	4000 Hz	10 V	$T_m = x_2 - x_1$ $= 0.00025s$ 4000 Hz
3.	Upper Sideband (USB)	-	-	10 V	$T_m = x_2 - x_1$ $= 0.000227s$ 4400 Hz
4.	Lower Sideband (LSB)	-	-	10 V	$T_m = x_2 - x_1$ $= 0.000278s$ 3600 Hz

Ex: 3
24/7/26

SIMULATION OF SINGLE - SIDEBAND (SSB) Modulation find its spectrum.

Aim:

To Simulate the Single Sideband Suppressed carrier (SSB-SC) modulation using Scilab and observe the message signal, carrier signal, upper sideband (USB) lower sideband (LSB)

APPARATUS REQUIRED :

- * A Laptop
- * Scilab software

CODE :

```
CIC ;  
clear ;  
close ;  
am = 4 ;  
fm = 400 ;  
ac = 8 ;  
fc = 4000 ;  
fs = 40000 ;  
t = 0 : 1/fs : 2/fm ;
```

Calculation:

1. Message Signal:

Practical time :

$$T_m = x_2 - x_1$$

$$= 0.0050 - 0.0025$$

$$= 0.0025$$

Frequency

$$f_m = \frac{1}{T_m}$$

$$= \frac{1}{0.0025}$$

$$= 400 \text{ Hz}$$

2. Carrier Signal:

Practical time :

$$T_m = x_2 - x_1$$

$$= 0.00028 - 0.00003$$

$$= 0.00025 \text{ s}$$

Frequency :

$$f_m = \frac{1}{T_m}$$

$$= \frac{1}{0.00025}$$

$$= 4000 \text{ Hz}$$

$$e_{m1} = a_m \cos(2 \times 10^4 \pi f_m t);$$

$$e_{m2} = a_m \cos(1.57 - (2 \times 10^4 \pi f_m t));$$

$$e_{c1} = a_c \cos(2 \times 10^6 \pi f_c t);$$

$$e_{c2} = a_c \cos(1.57 - (2 \times 10^6 \pi f_c t));$$

figure (1);

subplot (4, 1, 1);

plot (t, e_{m1});

xgrid ();

title ("Message Signal");

xlabel ("Time (s)");

ylabel ("Amplitude");

subplot (4, 1, 2);

plot (t, e_{c1});

xgrid ();

title ("Carrier Signal");

xlabel ("time (s)");

ylabel ("Amplitude");

$$e_{am1} = a_c \cdot (1.57 - (e_{m1} / a_c)) \cdot \cos(2 \times 10^6 \pi f_c t);$$

$$e_{am2} = a_c \cdot (1.57 + (e_{m1} / a_c)) \cdot \cos(2 \times 10^6 \pi f_c t);$$

$$edsb_{sc1} = e_{am1} - e_{am2};$$

$$e_{am3} = a_c * (1.57 + (e_{m2}/a_c))^* \sin(2 * \%pi * f_c * t);$$

$$e_{am4} = a_c * (1.57 - (e_{m1}/a_c))^* \sin(2 * \%pi * f_c * t);$$

$$e_{dsb_{sc}2} = e_{am3} - e_{am4};$$

$$e_{lsb} = e_{dsb_{sc}1} + e_{dsb_{sc}2};$$

$$e_{usb} = e_{dsb_{sc}1} - e_{dsb_{sc}2};$$

subplot (4, 1, 3);

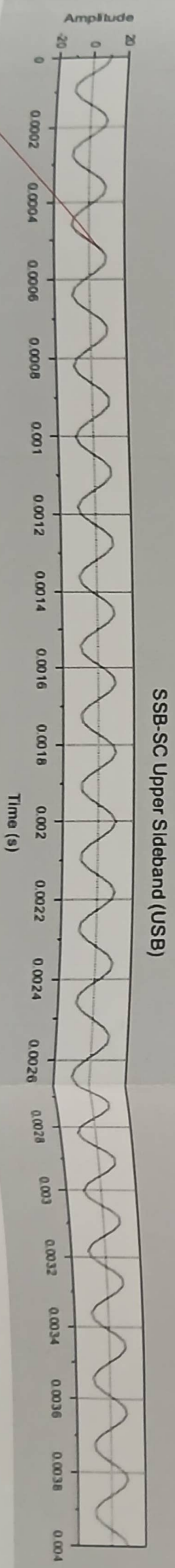
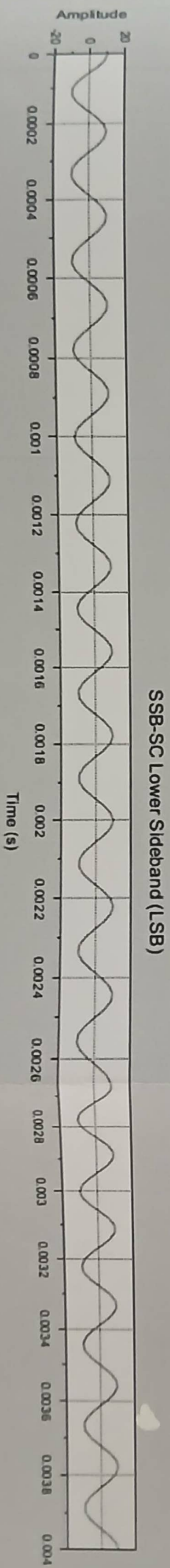
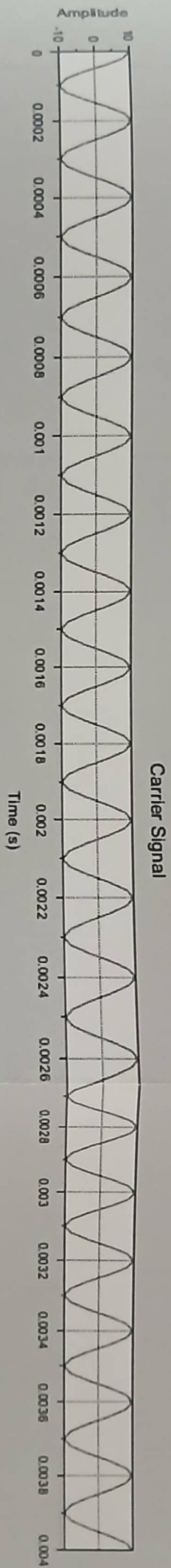
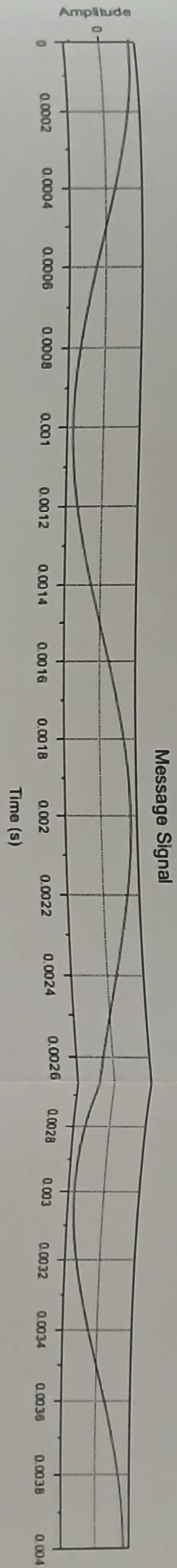
plot (t, e_{lsb});

x grid ();

title ("SSB - sc Lower Sideband (LSB)");

x label ("time (s)");

y label ("Amplitude");



Scilab program implementation (20)	Wave form tabulation (20)	Calculation & Result (20)	Viva voice (20)	Result Record Observation (20)	Total mark
20	20	20	15	20	75

RESULT:

Thus the Simulation of Single Side band (SSB) modulation was successfully generated using Scilab. The message signal and Carrier²⁰ signal were observed.